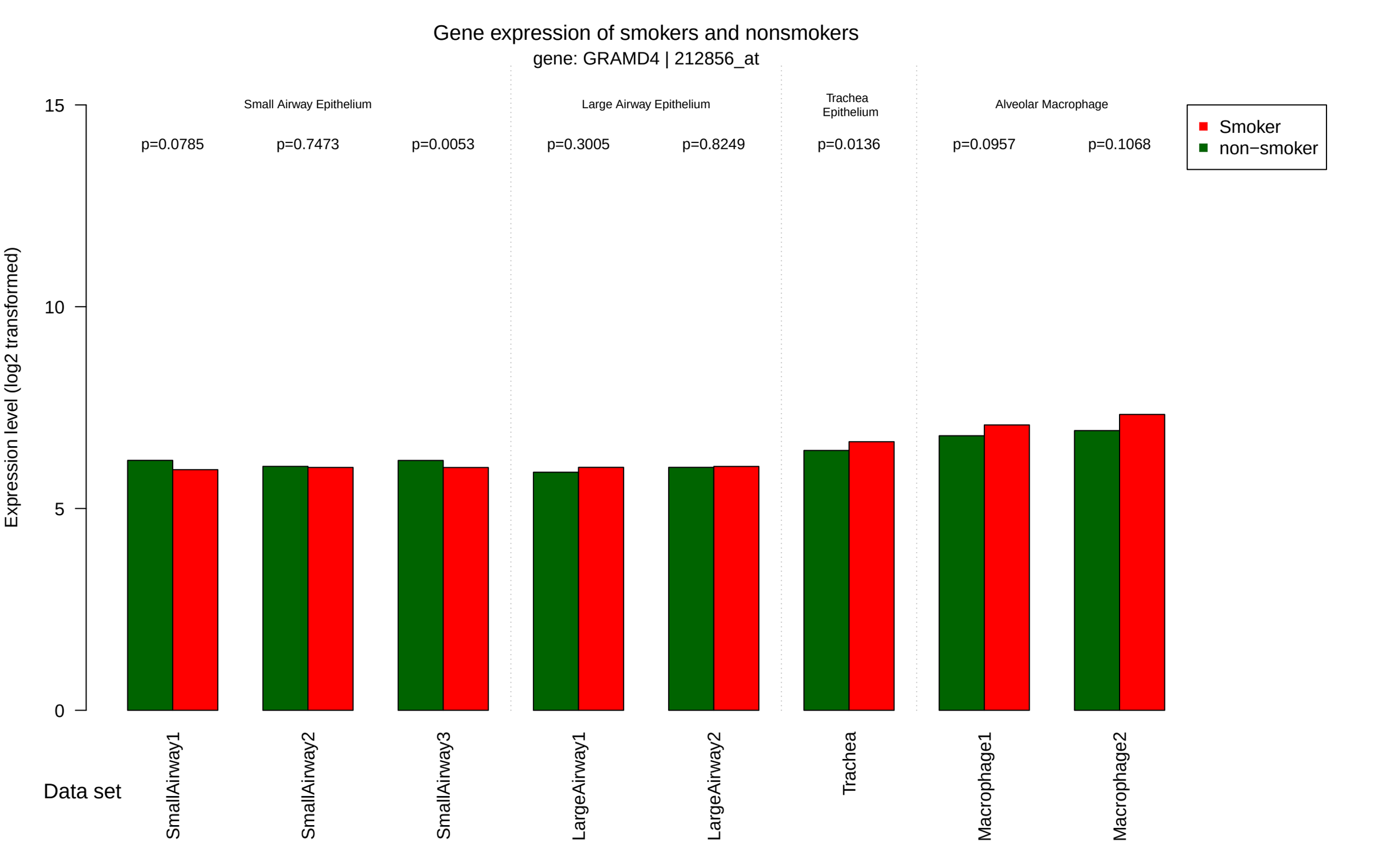
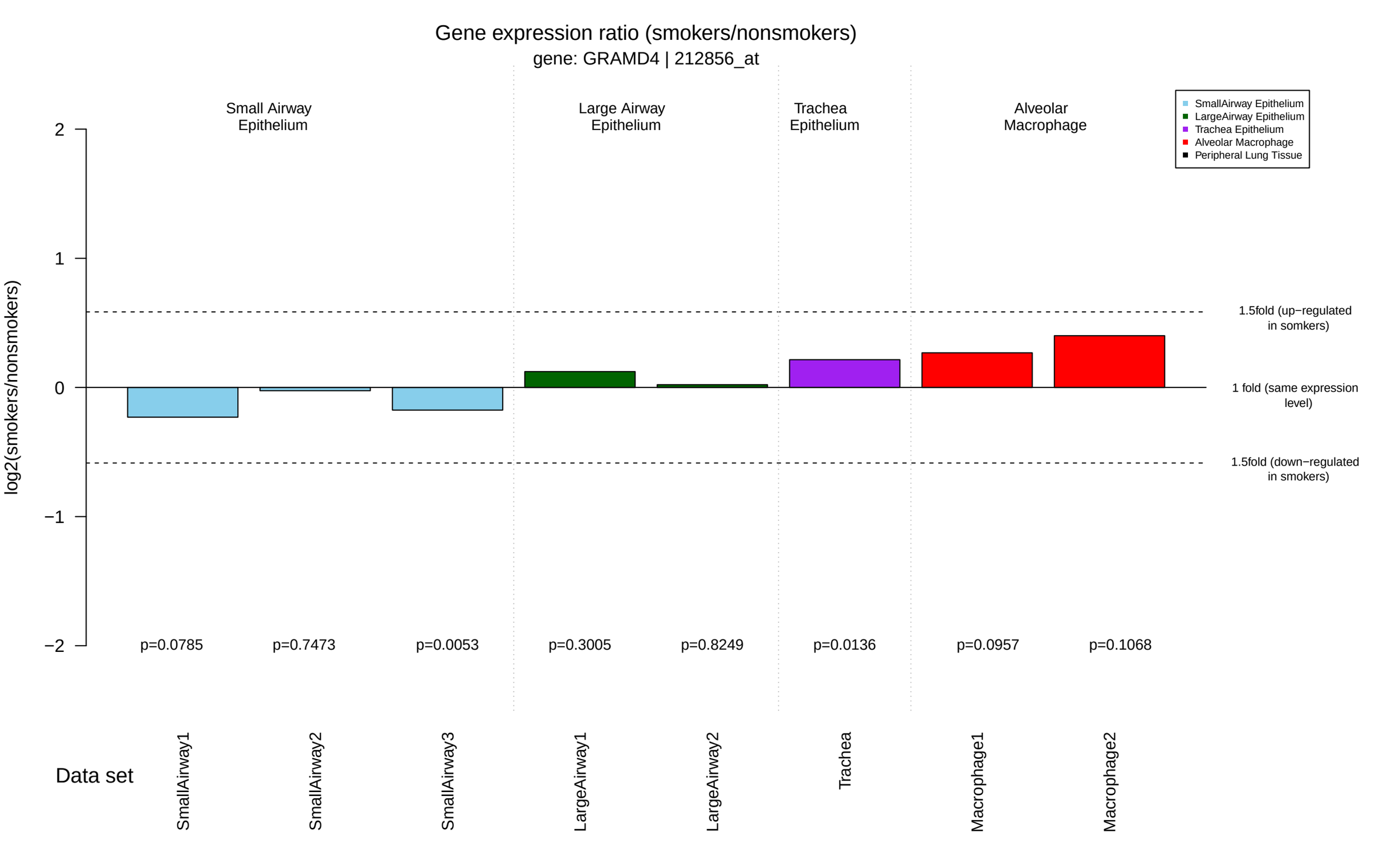
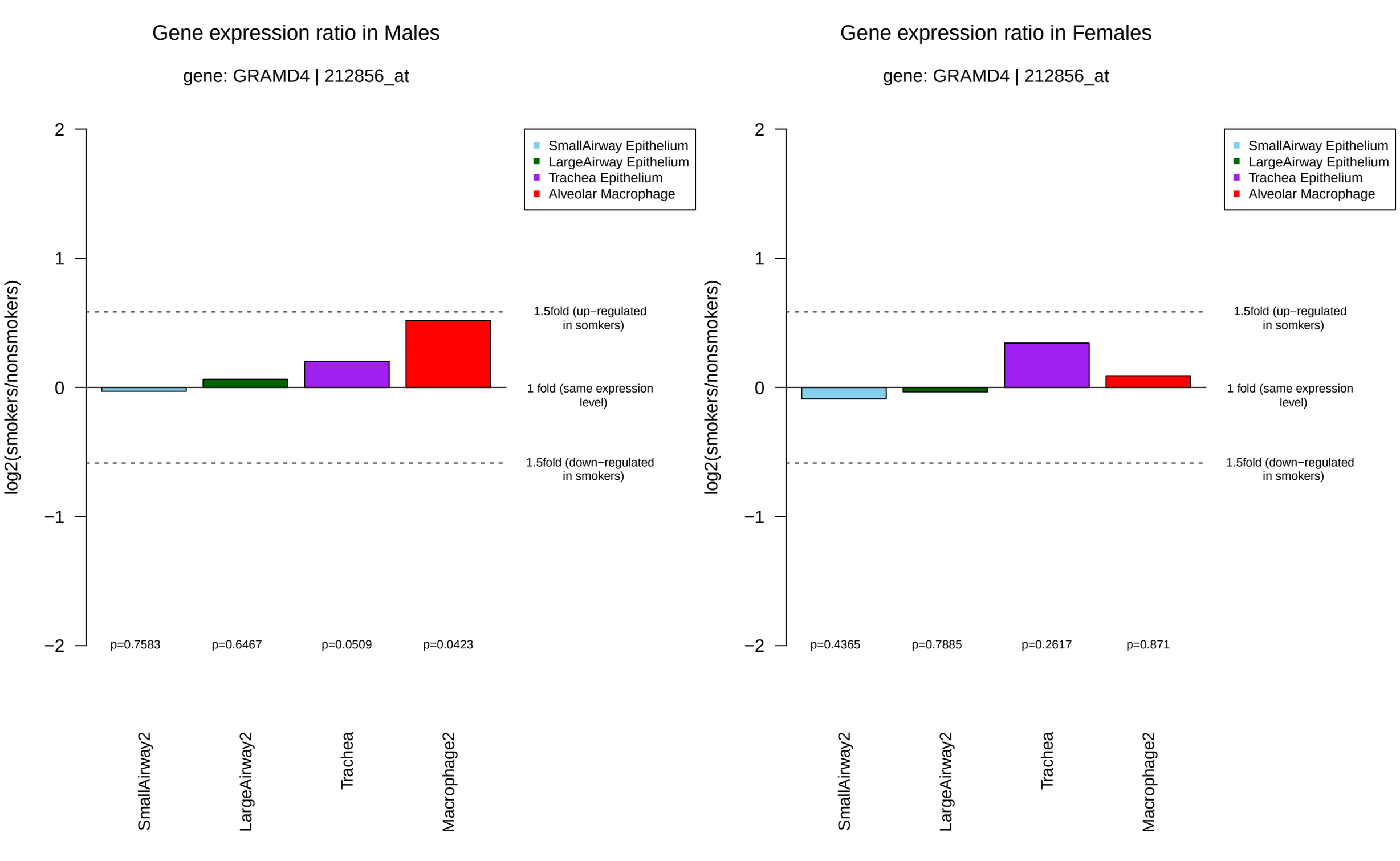
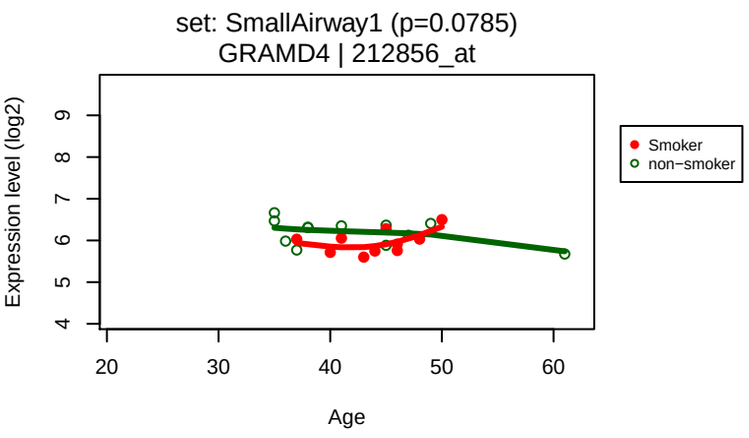
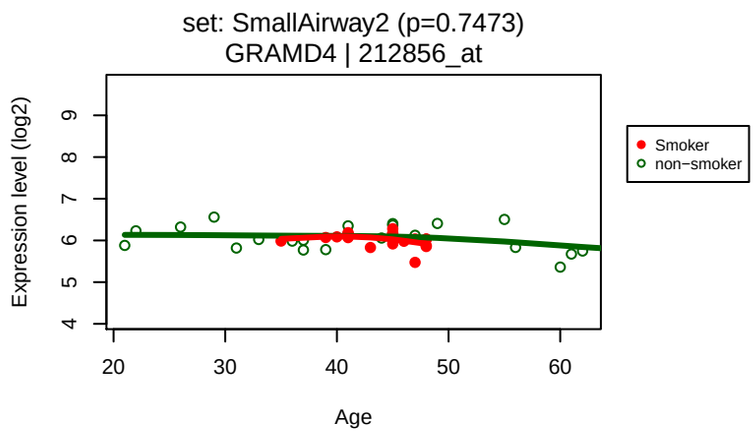
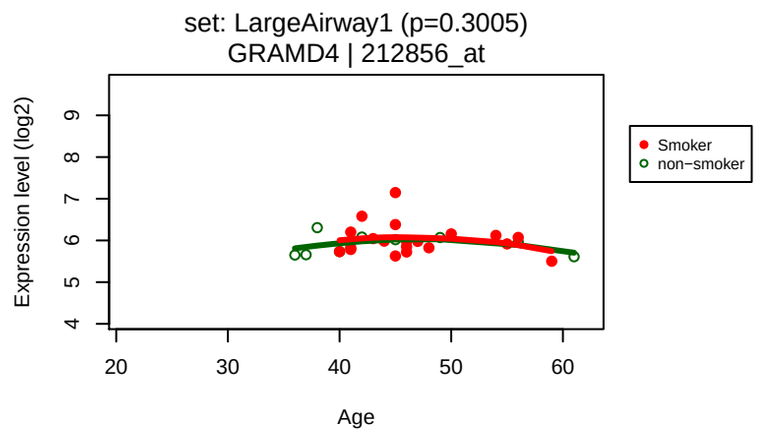

Gene expression of smokers and nonsmokers
gene: GRAMD4 | 212856_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0785
p=0.7473
p=0.0053
p=0.3005
p=0.8249
p=0.0136
p=0.0957
p=0.1068
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

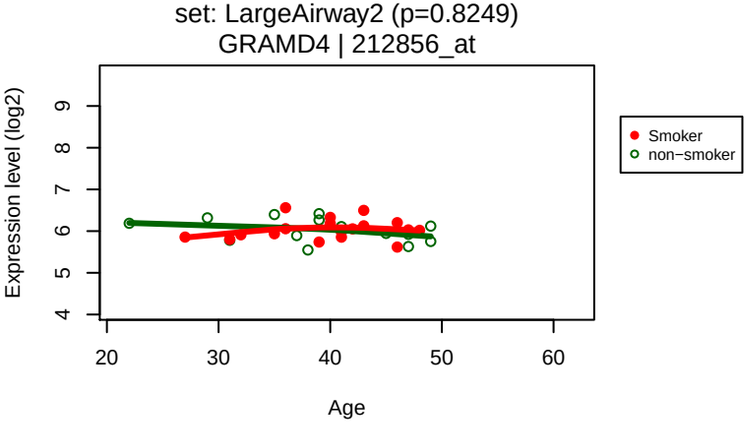

Gene expression ratio (smokers/nonsmokers)
gene: GRAMD4 | 212856_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0785
p=0.7473
p=0.0053
p=0.3005
p=0.8249
p=0.0136
p=0.0957
p=0.1068
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

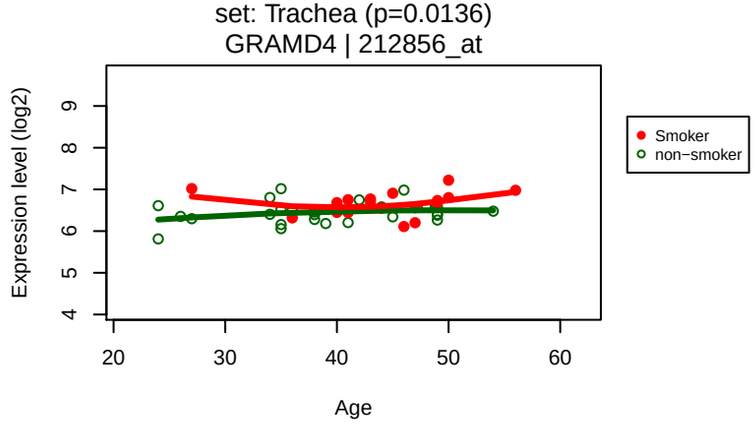

Gene expression ratio in Males
gene: GRAMD4 | 212856_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7583
p=0.6467
p=0.0509
p=0.0423
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRAMD4 | 212856_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4365
p=0.7885
p=0.2617
p=0.871
SmallAirway2
LargeAirway2
Trachea
Macrophage2

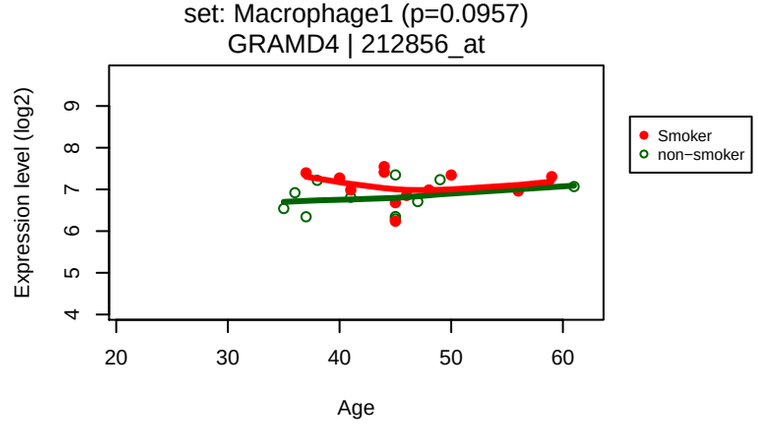

set: SmallAirway1 (p=0.0785)
GRAMD4 | 212856_at
Expression level (log2)
Age
Smoker
non−smoker

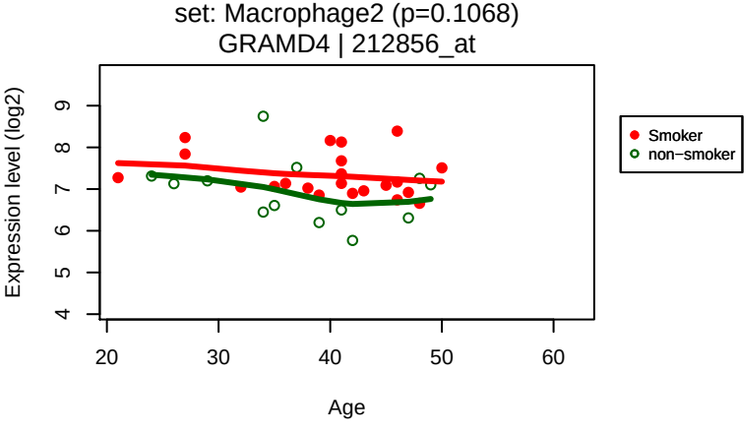

set: SmallAirway2 (p=0.7473)
GRAMD4 | 212856_at
Expression level (log2)
Age
Smoker
non−smoker

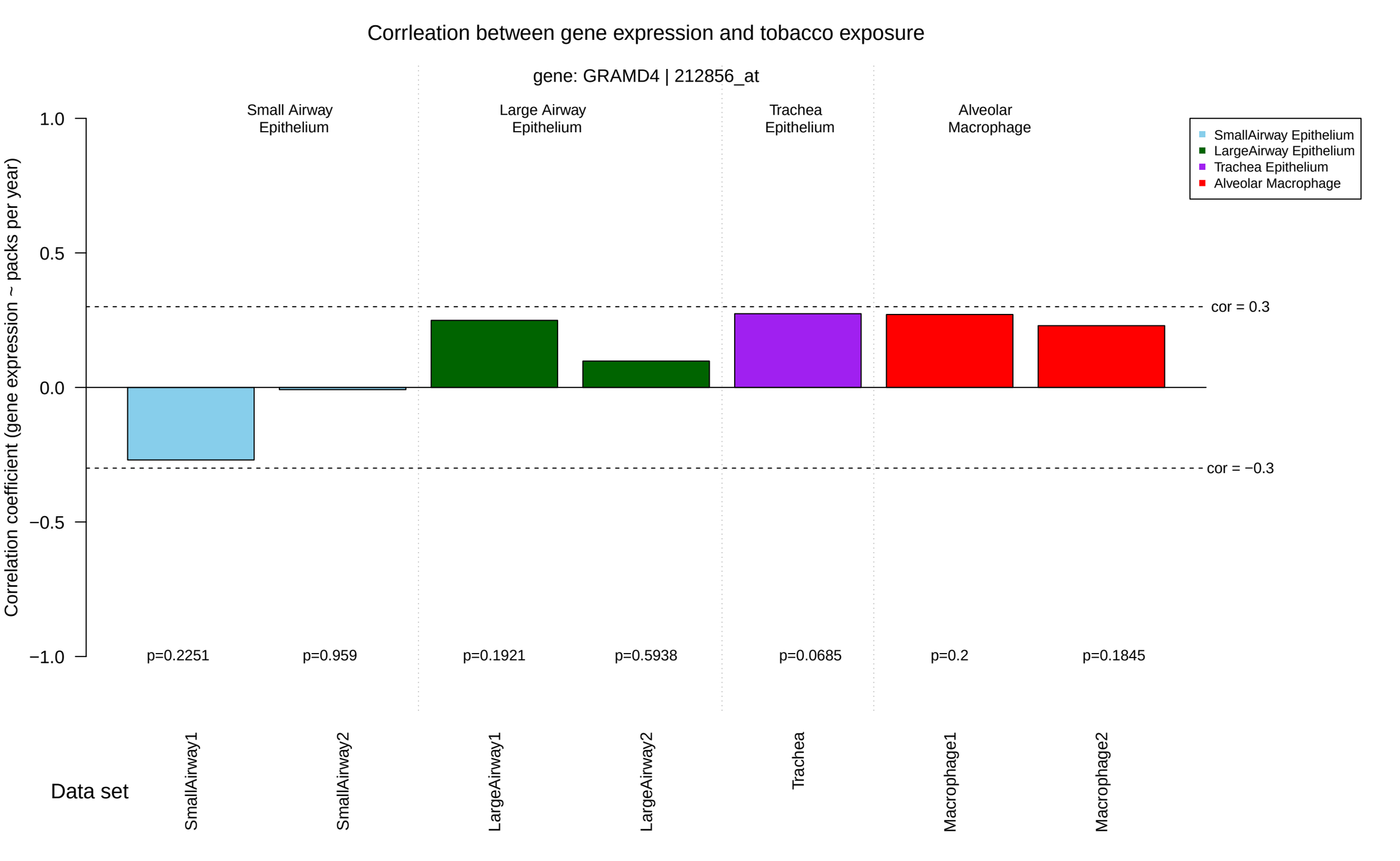

set: LargeAirway1 (p=0.3005)
GRAMD4 | 212856_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8249)
GRAMD4 | 212856_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0136)
GRAMD4 | 212856_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0957)
GRAMD4 | 212856_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1068)
GRAMD4 | 212856_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRAMD4 | 212856_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2251
p=0.959
p=0.1921
p=0.5938
p=0.0685
p=0.2
p=0.1845
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2